

Dongwook Kwon | AI Engineer ■

Agentic AI Systems

11F, 37, Geumto-ro 80beon-gil, Sujeong-gu, Seongnam-si
Gyeonggi-do, 13453, South Korea

☎ +82 10-6612-9246 • ✉ dwkwon@suresofttech.com • 🌐 dongwooky.github.io
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Experience

Suresoft Technologies

Seongnam-si, Korea

AI Engineer, Intelligent Agent Technology Team

Dec. 2025 – Present

- Developing an AI validation system utilizing AI agent-based architectures for enterprise model verification.
- Designing autonomous and collaborative AI agents that verify and evaluate machine-learning models at scale.
- Engineered and delivered a critical agent-based codebase for a mission-critical vehicle control data project for Hyundai Motor Company.

Bio-Computing and Machine Learning Laboratory, Kwangwoon University

Seoul, Korea

Research Assistant

Aug. 2021 – Dec. 2025

- Researched deep-learning algorithms for anomaly detection in complex time-series data, focusing on critical cyber-physical systems including nuclear power generation environments.
- Co-developed an Active Kill-Switch and Biomarker-Based Defense System for life-threatening IoT medical devices.
- Published in EMNLP 2025 (Oral) and ACL 2026; co-author on two granted Republic of Korea patents.

LG Electronics Canada (with University of Toronto)

Toronto, ON, Canada

Visiting Researcher

Jan. 2025 – Jul. 2025

- Built an LLM-driven assessment framework for evaluating performance of agentic AI systems.
- Work fed directly into peer-reviewed publications at EMNLP 2025 and ACL 2026.

Qualcomm Institute, University of California San Diego

San Diego, CA, USA

AI Development Project Participant, Summer Program

Jul. 2022 – Aug. 2022

- Built a personality-based drug-addiction prediction system using machine-learning models.
- Recognized with the program's Achievement Award.

Education

Kwangwoon University

Seoul, Korea

M.S. in Computer Engineering, GPA: 4.33/4.5 (98.1%)

Mar. 2024 – Feb. 2026

University of Toronto

Toronto, ON, Canada

Graduate Research Program, GPA: 3.68/4.0

Jan. 2025 – Jul. 2025

Kwangwoon University

Seoul, Korea

B.S. in Electronics and Communications Engineering, GPA: 3.41/4.5 (88.1%)

Mar. 2019 – Feb. 2024

Honors & Awards

Major Awards

Aug. 2022: Achievement Award, Qualcomm Institute AI Development Project — UC San Diego *United States*

Dec. 2021: Gold Award (Ministerial Award), 2021 Smart Maritime Logistics Project — Minister of Oceans and Fisheries *South Korea*

Academic Recognition

Nov. 2025: Outstanding Student Paper Award, Conference of the Institute of Electronics and Information Engineers *South Korea*

Nov. 2024: Best Poster Award (Silver), 10th International Biomedical Engineering Conference (IBEC 2024) *South Korea*

Nov. 2023: Outstanding Paper Award, Conference of the Korean Society of Medical and Biological Engineering *South Korea*

Competitions

Dec. 2025: Silver Award, 23rd Embedded Software Competition (KIISE) *South Korea*

Dec. 2022: Bronze Award, Hanium ICT Mentoring Competition — FKII *South Korea*

Sep. 2022: Excellence Award, 18th Kwangwoon ICT Exhibition (KWIX) — Kwangwoon University *South Korea*

Oct. 2021: Grand Prize, MY (Multi-Y) Capstone Design Competition — Kwangwoon University *South Korea*

Funded Research Projects

LG Electronics Canada

Agentic AI Evaluation Framework

Developed an evaluation framework for agentic AI systems using LLMs to assess performance.

Toronto, ON, Canada

Jan. 2025 – Present

Sejong University

AI Intrusion Detection for APR1400

Developed an AI defense system to protect nuclear power plants from cyber attacks.

Seoul, Korea

Jul. 2022 – Present

Ministry of Science and ICT

Quadruped Robot with Vision SLAM

Developed an autonomous quadruped robot using a navigation algorithm with SLAM and a depth camera.

Seoul, Korea

Apr. 2022 – Nov. 2022

Qualcomm Institute, University of California San Diego

Personality-Based Drug Addiction Prediction System

Developed a personality-based drug addiction prediction system using a machine learning model.

San Diego, CA, USA

Jul. 2022 – Aug. 2022

Kookmin University

Active Kill-Switch and Biomarker-Based Defense for IoT Medical Devices

Developed a security system for heart implant devices using an anomaly detection algorithm.

Seoul, Korea

Sep. 2021 – Dec. 2022

Publications

Journal Articles

2025: Y. Nam, J. Lee, H. Lee, **D. Kwon**, M. Yeo, S. Kim, R. Sohn, and C. Park, "Designing a remote photoplethysmography-based heart rate estimation algorithm during a treadmill exercise," *Electronics*, vol. 14,

no. 5, p. 890, 2025.

2025: D. Kwon, Y. Kang, J. Lee, Y. Nam, J. Won, K. K. Kim, D. D. Kim, and C. Park, “Graph-Enhanced Bidirectional GRU for Anomaly Detection in Power Generation Environments,” under review at *Data Mining and Knowledge Discovery (Springer Nature)*, 2025.

Conference Papers.....

2026: R. Chang[†], **D. Kwon**[†], J. Lee[†], and N. Verma, “CascadeDebate: Multi-Agent Deliberation for Cost-Aware LLM Cascades,” in *Proc. 64th Annu. Meeting Assoc. Comput. Linguistics (ACL) — Industry Track (Poster)*, 2026. [†]Equal contribution.

2025: J. Lee[†], R. Chang[†], **D. Kwon**[†], H. Singh, and N. Verma, “GEMMAS: Graph-based Evaluation Metrics for Multi Agent Systems,” in *Proceedings of the EMNLP 2025 Industry Track (Oral)*, 2025. [†]Equal contribution.

2024: **D. Kwon**, M. Kim, Y. Kang, J. Lee, and C. Park, “Hybrid neural network model for anomaly detection in implantable devices using graph attention networks and transformers,” in *Proc. 10th Int. Biomed. Eng. Conf. (IBEC)*, 2024. **Best Poster Award — Silver.**

2023: **D. Kwon**, Y. Kang, and C. Park, “Enhancing medical device security with GNN-GRU anomaly detection model,” in *Proc. 62nd KOSOMBE Autumn Conf.*, pp. 204–205, 2023. **Outstanding Paper Award.**

Patents

Apr 2026: Issued Patent, “Method for Implantable Medical Device to Detect Anomaly in Real Time,” KR 10-2961984.

Aug 2025: Issued Patent, “GPT API-Based Network Anomaly Detection,” KR 10-2848814.

Aug 2024: Patent Application, “Categorical Data of Networks Based Network Anomaly Detection Apparatus and Method Thereof,” KR Patent App. No. 10-2024-0110559.

Leadership Experience

President, Kwangwoon International Student Association — Lead intercultural programs and student engagement initiatives.

Director of Filming and Editing, Kwangwoon Broadcasting Center (KWBC) — Oversaw media production and content delivery.

Class Representative, Dept. of Electronics and Communications Engineering — Represented student interests in academic affairs.

Executive Member, Amateur Radio Club — Organized radio workshops and technical exhibitions.